SHWETA KUMARI

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Professional Summary

Final year Computer Science Undergraduate with hands-on experience in full-stack development, AI-driven applications, and data analytics. Skilled at designing scalable systems, building responsive UIs, and integrating APIs. Experienced in team-based development and contributing to open-source projects, with a strong focus on delivering user-focused, practical solutions.

Education

Bangalore Institute of Technology, Bengaluru, India | *Computer Science Engineering | CGPA:* ***9.24*** **2023 -2027** *(expected)*

Shivam Convent, Patna, Bihar, India, *AISSCE (Class XII) | Aggregate:92.6%* **2021 -2022**

BD Public School, Patna, Bihar, India, *AISSE (Class X) | Aggregate:98.2%* **2019 -2020**

Skills

Programming: C, Python, Java | **Web Technologies:** HTML, CSS, JavaScript, React, Tailwind CSS | **Databases:** DBMS, SQL |
Core Concepts: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, Software Engineering |
Tools & Platforms: Git, Figma, Streamlit

Projects

Crimora- *Crime Reporting and Analytical* Platform **Live Demo**

- Developed a secure web platform for crime data reporting, visualization, and predictive analytics.
- Enforced user authentication, session logging, and dashboards to detect predictive trends and identify hotspots
- Designed and normalized a relational database schema (3NF) for efficient storage and retrieval of data.

Moods - *Mood-based Entertainment Recommendation* Platform **Live Demo**

- Developed a full-stack web application with React frontend and Python FastAPI backend, implementing 3 core API endpoints for mood analysis and content recommendations.
- Integrated OpenAI GPT-4 API for natural language processing, enabling real-time mood detection across 10+ emotion categories.
- Designed a responsive user interface using Tailwind CSS with 10+ interactive mood selection options, dynamic content filtering, and data handling for a seamless user experience.
- Applied NoSQL database architecture using MongoDB and Motor async driver, storing structured user interaction data with proper error handling and fallback mechanisms.

BehaviorDrift - *ML-Based GitHub Repository Anomaly Detection Platform* **Live Demo**

- Architected a full-stack anomaly detection system (Python, FastAPI, React/ Vite, Tailwind) with a modular pipeline architecture separating data collection, feature engineering, ML inference, and reporting into independently testable components.
- Designed and implemented a rolling-baseline ML pipeline using scikit-learn's Isolation Forest, engineering a 4-dimensional feature vector (commit velocity, author diversity, dependency churn, commits-per-author) per time window to prevent data leakage across temporal splits.
- Built a resilient 3-tier LLM integration (Claude → Gemini → deterministic template fallback) for automated report generation, ensuring the system degrades gracefully if an external API is unavailable.
- Implemented a pluggable persistence layer supporting both a production vector database (Qdrant) and a local JSON fallback, enabling the same codebase to run fully offline for development/testing or connect to live infrastructure in production.
- Achieved high test coverage across 7 core modules with a fully mocked pytest suite (GitHub API and LLM calls mocked via requests-mock), enabling deterministic, dependency-free CI pipelines.

Academic and Extracurricular Achievements

- Awarded the Constellation Prize at Ideaverse 2025 for presenting a group prototype of an AI-powered medical diagnosis tool, Aroviaa.
- Contributed to open-source projects, focusing on frontend enhancements and minor feature improvements in GirlScript Summer of Code (GSSOC) as a Contributor, 2024 & 2025.
- Prepared technical content, event summaries, and write-ups to support community-driven workshops as a Content Writer for SDI (Student Developers' Initiative), the official CSE department club.
- Served as a Creative Team Member at TEDx BIT Bangalore (May 2024 – May 2025), assisting in event planning, branding, and visual content creation for a successfully organized TED event.